

The Complex Orthogonal Measure and the Spectral Representation of the Hardy Z -Function

Abstract

Let θ be the Riemann–Siegel theta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z -function, and $H(x) := Z(\theta^{-1}(x))/\sqrt{\theta'(\theta^{-1}(x))}$ the warped Hardy signal. Given as input the companion results

- Theorem A of *Three Missing Theorems*, which establishes

$$C(\eta) := \lim_{X \rightarrow \infty} \frac{1}{X - X_0} \int_{X_0}^{X-|\eta|} H(x) \overline{H(x+|\eta|)} dx = 2 \cos |\eta|, \quad \eta \in \mathbb{R},$$

with the quantitative rate $|C_X(\eta) - C_{\text{diag}}(\eta)| \leq 8\pi(1 + |\eta|) \log^3(X/X_0)/(X - X_0) + 0.349/\sqrt{T_0}$;

- the Bochner factorization $C(\eta) = \int_{\mathbb{R}} e^{i\eta\xi} d\nu(\xi)$ with $\nu = \delta_{-1} + \delta_{+1}$ from *Existence of the Spectral Measure in the Limit*,

we define the complex orthogonal measure Φ of H by an *explicit Cramér-type pre-limit formula*, prove convergence of the pre-limit in the time-average Hilbert space generated by the shifts of H , and verify by Fubini the spectral representation

$$H(y) = \int_{\mathbb{R}} e^{i\xi y} d\Phi(\xi), \quad y \in \mathbb{R},$$

as an identity in that Hilbert space. The measure Φ is Hilbert-space-valued, orthogonally-scattered on the Borel σ -algebra of $\mathbb{R}$, supported on $\{-1, +1\}$, with intensity $\langle \Phi(A), \Phi(B) \rangle = \nu(A \cap B)$. The corresponding representation of the Hardy Z -function is

$$Z(t) = \sqrt{\theta'(t)} \int_{\mathbb{R}} e^{i\xi\theta(t)} d\Phi(\xi)$$

in the same Hilbert space.

1 Inputs and notation

Fix $T_0 \geq 200$ and set $X_0 := \theta(T_0)$. For $X > X_0$ set $T := \theta^{-1}(X)$. The Riemann–Siegel theta function $\theta : [T_0, \infty) \rightarrow [X_0, \infty)$ is a diffeomorphism with $\theta'(t) > 0$, and $H \in C_{\text{loc}}^\infty([X_0, \infty); \mathbb{C})$ is locally bounded.

Time-average inner product. For $f, g : [X_0, \infty) \rightarrow \mathbb{C}$ bounded and measurable, write, whenever the limit exists,

$$\langle f, g \rangle_{\sharp} := \lim_{X \rightarrow \infty} \frac{1}{X - X_0} \int_{X_0}^X f(x) \overline{g(x)} dx.$$

Theorem A states exactly that the limit $\langle H_s, H_t \rangle_{\#}$ exists for every $s, t \in \mathbb{R}$, where

$$H_s(x) := H(x + s) \cdot \mathbf{1}_{\{x+s \geq X_0\}},$$

and equals $C(s - t) = 2 \cos(s - t) = e^{-i(s-t)} + e^{+i(s-t)}$. (The indicator does not affect the time average, which is insensitive to a compact perturbation of the integrand.)

Pre-Hilbert space. Let $\mathcal{E}$ denote the linear span of $\{H_s : s \in \mathbb{R}\}$ over $\mathbb{C}$. Define the sesquilinear form on $\mathcal{E}$ by

$$\left\langle \sum_k \alpha_k H_{s_k}, \sum_j \beta_j H_{t_j} \right\rangle_{\#} := \sum_{k,j} \alpha_k \overline{\beta_j} C(s_k - t_j).$$

This is well-defined because each individual time average $\langle H_{s_k}, H_{t_j} \rangle_{\#}$ exists by Theorem A and is bilinear in the coefficients. The Gram matrix $[C(s_k - t_j)]_{k,j}$ is Hermitian and positive semidefinite: by the Bochner factorization $\nu = \delta_{-1} + \delta_{+1}$,

$$\sum_{k,j} \alpha_k \overline{\alpha_j} C(s_k - s_j) = \int_{\mathbb{R}} \left| \sum_k \alpha_k e^{i\xi s_k} \right|^2 d\nu(\xi) = \left| \sum_k \alpha_k e^{-is_k} \right|^2 + \left| \sum_k \alpha_k e^{+is_k} \right|^2 \geq 0.$$

Quotient $\mathcal{E}$ by its null space $\mathcal{N} = \{f \in \mathcal{E} : \langle f, f \rangle_{\#} = 0\}$ and complete in the resulting norm. Call the resulting Hilbert space $\mathcal{H}$. The natural injection $\iota : \mathcal{E}/\mathcal{N} \hookrightarrow \mathcal{H}$ has dense image.

Write $\widetilde{H}_s \in \mathcal{H}$ for the image of $H_s \in \mathcal{E}$ under the composition $\mathcal{E} \rightarrow \mathcal{E}/\mathcal{N} \hookrightarrow \mathcal{H}$. Then

$$\langle \widetilde{H}_s, \widetilde{H}_t \rangle_{\mathcal{H}} = C(s - t) = e^{-i(s-t)} + e^{+i(s-t)}.$$

Extension of the inner product to Bochner integrals of $\mathcal{H}$ -valued functions. For a continuous compactly supported function $K : [X_0, \infty) \rightarrow \mathbb{C}$, the $\mathcal{H}$ -valued integral

$$\int_{X_0}^{\infty} K(x) \widetilde{H}_x dx \in \mathcal{H}$$

is defined as the Bochner integral of the continuous $\mathcal{H}$ -valued map $x \mapsto K(x) \widetilde{H}_x$. (Continuity in x in the $\mathcal{H}$ -norm is implied by $\|\widetilde{H}_{x+h} - \widetilde{H}_x\|_{\mathcal{H}}^2 = C(0) - 2 \operatorname{Re} C(h) + C(0) = 4 - 4 \cos h \rightarrow 0$ as $h \rightarrow 0$.) Its squared norm satisfies the usual Bochner identity

$$\left\| \int K(x) \widetilde{H}_x dx \right\|_{\mathcal{H}}^2 = \iint K(x) \overline{K(y)} C(x - y) dx dy.$$

2 The explicit pre-limit Cramér formula

Definition 1 (Pre-limit band coefficients). For $X > X_0$ set

$$\Phi_{+,X} := \frac{1}{X - X_0} \int_{X_0}^X e^{-ix} \widetilde{H}_x dx \in \mathcal{H}, \quad \Phi_{-,X} := \frac{1}{X - X_0} \int_{X_0}^X e^{+ix} \widetilde{H}_x dx \in \mathcal{H}.$$

The two integrals are Bochner integrals of the continuous $\mathcal{H}$ -valued integrands $x \mapsto e^{\mp ix} \widetilde{H}_x$, as in the paragraph closing Section 1. They are explicitly given by a finite Cesaro average in $\mathcal{H}$ and are defined for every $X > X_0$, without passing to any limit.

Definition 2 (Pre-limit orthogonal measure). For every Borel set $A \subset \mathbb{R}$ and every $X > X_0$ define

$$\Phi_X(A) := \mathbf{1}_A(+1) \Phi_{+,X} + \mathbf{1}_A(-1) \Phi_{-,X} \in \mathcal{H}.$$

$\Phi_X(\cdot)$ is a finitely supported $\mathcal{H}$ -valued set function carried on $\{-1, +1\}$. Countable additivity in A is immediate from the two-atom form. The dependence on X is entirely in the two coefficients $\Phi_{\pm,X}$.

Remark 3. Definition 1 is the Cramér pre-limit formula specialised to the spectrum $\{\pm 1\}$ established by Theorem A. For a general weakly harmonizable integrand with spectral measure ν on $\mathbb{R}$ and shift family $\widetilde{H}_x$, the Cramér pre-limit for a Borel set A takes the form

$$\Phi_X(A) = \frac{1}{X - X_0} \int_{X_0}^X \widehat{\mathbf{1}}_A(x) \widetilde{H}_x dx,$$

understood in a suitable regularised sense when $\widehat{\mathbf{1}}_A$ is not a tempered function. In the present case $\text{supp } \nu = \{\pm 1\}$, so only the values of $\widehat{\mathbf{1}}_A$ at the two atoms contribute: $\widehat{\mathbf{1}}_A(x)$ is replaced by $\mathbf{1}_A(+1)e^{-ix} + \mathbf{1}_A(-1)e^{+ix}$, which is the formula in Definition 1.

3 Cauchy convergence of the pre-limit in $\mathcal{H}$

Lemma 4 (Inner products of the pre-limit band coefficients). *For every $X, X' > X_0$,*

$$\begin{aligned} \langle \Phi_{+,X}, \widetilde{H}_y \rangle_{\mathcal{H}} &= \frac{1}{X - X_0} \int_{X_0}^X e^{-ix} C(x - y) dx = e^{-iy} + R_+(X; y), \\ \langle \Phi_{-,X}, \widetilde{H}_y \rangle_{\mathcal{H}} &= \frac{1}{X - X_0} \int_{X_0}^X e^{+ix} C(x - y) dx = e^{+iy} + R_-(X; y), \end{aligned}$$

where

$$R_+(X; y) = \frac{e^{iy}}{X - X_0} \int_{X_0}^X e^{-2ix} dx = \frac{e^{iy}}{-2i(X - X_0)} (e^{-2iX} - e^{-2iX_0}),$$

and R_- is its complex conjugate expression with the roles of the two exponents interchanged. In particular $|R_{\pm}(X; y)| \leq 1/(X - X_0)$.

Proof. By the definition of the Bochner integral, $\langle \int K(x) \widetilde{H}_x dx, \widetilde{H}_y \rangle_{\mathcal{H}} = \int K(x) \langle \widetilde{H}_x, \widetilde{H}_y \rangle_{\mathcal{H}} dx = \int K(x) C(x - y) dx$. Take $K(x) = e^{-ix}/(X - X_0)$ on $[X_0, X]$ and zero elsewhere. Since $C(x - y) = e^{-i(x-y)} + e^{+i(x-y)}$,

$$\int_{X_0}^X e^{-ix} C(x - y) dx = e^{iy} \int_{X_0}^X e^{-2ix} dx + e^{-iy} \int_{X_0}^X 1 dx.$$

Dividing by $X - X_0$ gives $e^{-iy} + R_+(X; y)$ with R_+ as stated. The oscillatory integral $\int_{X_0}^X e^{-2ix} dx$ is bounded in modulus by 1, giving $|R_+(X; y)| \leq 1/(X - X_0)$. The computation for $\Phi_{-,X}$ is identical with $-i$ replaced by $+i$ in the kernel. $\square$

Lemma 5 (Inner products of the pre-limit band coefficients among themselves). *For every $X, X' > X_0$,*

$$\begin{aligned} \langle \Phi_{+,X}, \Phi_{+,X'} \rangle_{\mathcal{H}} &= 1 + E_{++}(X, X'), \\ \langle \Phi_{-,X}, \Phi_{-,X'} \rangle_{\mathcal{H}} &= 1 + E_{--}(X, X'), \\ \langle \Phi_{+,X}, \Phi_{-,X'} \rangle_{\mathcal{H}} &= E_{+-}(X, X'), \end{aligned}$$

with $|E_{..}(X, X')| \leq 2/\min(X - X_0, X' - X_0)$.

Proof. For the first inner product, using the Bochner identity,

$$\begin{aligned}
\langle \Phi_{+,X}, \Phi_{+,X'} \rangle_{\mathcal{H}} &= \frac{1}{(X - X_0)(X' - X_0)} \int_{X_0}^X \int_{X_0}^{X'} e^{-ix} e^{iy} C(x - y) dx dy \\
&= \frac{1}{(X - X_0)(X' - X_0)} \int_{X_0}^X \int_{X_0}^{X'} e^{-ix+iy} [e^{-i(x-y)} + e^{+i(x-y)}] dx dy \\
&= \frac{1}{(X - X_0)(X' - X_0)} \int_{X_0}^X \int_{X_0}^{X'} [e^{-2ix+2iy} + 1] dx dy \\
&= 1 + \frac{1}{(X - X_0)(X' - X_0)} \int_{X_0}^X e^{-2ix} dx \cdot \int_{X_0}^{X'} e^{+2iy} dy.
\end{aligned}$$

Each one-variable oscillatory integral has modulus bounded by 1, so the product of the two quotients has modulus bounded by $1/((X - X_0)(X' - X_0))$. Absorbing this into the stated bound $2/\min(X - X_0, X' - X_0)$ (valid whenever $X - X_0 \geq 1$ or $X' - X_0 \geq 1$) gives the claim. The other two computations are identical with signs interchanged; for the cross term,

$$\langle \Phi_{+,X}, \Phi_{-,X'} \rangle_{\mathcal{H}} = \frac{1}{(X - X_0)(X' - X_0)} \int_{X_0}^X \int_{X_0}^{X'} [e^{-2ix} + e^{+2iy}] dx dy,$$

which is bounded in modulus by the same quantity. $\square$

Proposition 6 (Cauchy convergence). *The families $(\Phi_{+,X})_{X > X_0}$ and $(\Phi_{-,X})_{X > X_0}$ are Cauchy in $\mathcal{H}$ as $X \rightarrow \infty$. Their limits, denoted $\Phi_+ := \lim \Phi_{+,X}$ and $\Phi_- := \lim \Phi_{-,X}$, exist in $\mathcal{H}$ and satisfy*

$$\langle \Phi_+, \Phi_+ \rangle_{\mathcal{H}} = \langle \Phi_-, \Phi_- \rangle_{\mathcal{H}} = 1, \quad \langle \Phi_+, \Phi_- \rangle_{\mathcal{H}} = 0.$$

Proof. By Lemma 5,

$$\|\Phi_{+,X} - \Phi_{+,X'}\|_{\mathcal{H}}^2 = \langle \Phi_{+,X}, \Phi_{+,X} \rangle + \langle \Phi_{+,X'}, \Phi_{+,X'} \rangle - 2 \operatorname{Re} \langle \Phi_{+,X}, \Phi_{+,X'} \rangle \rightarrow 1 + 1 - 2 \cdot 1 = 0$$

as $\min(X, X') \rightarrow \infty$, with the explicit rate $4/\min(X - X_0, X' - X_0)$. By completeness of $\mathcal{H}$ the limit Φ_+ exists. The limit of $\langle \Phi_{+,X}, \Phi_{+,X'} \rangle$ as both $X, X' \rightarrow \infty$ is 1, so $\|\Phi_+\|_{\mathcal{H}}^2 = 1$. The analogous argument gives Φ_- and $\|\Phi_-\|_{\mathcal{H}}^2 = 1$. For orthogonality, by Lemma 5 $\langle \Phi_{+,X}, \Phi_{-,X'} \rangle_{\mathcal{H}} \rightarrow 0$ as $\min(X, X') \rightarrow \infty$, and continuity of the inner product on $\mathcal{H}$ transfers this to $\langle \Phi_+, \Phi_- \rangle_{\mathcal{H}} = 0$. $\square$

Definition 7 (The complex orthogonal measure). For every Borel set $A \subset \mathbb{R}$ set

$$\Phi(A) := \mathbf{1}_A(+1) \Phi_+ + \mathbf{1}_A(-1) \Phi_- \in \mathcal{H}.$$

Equivalently, $\Phi(A) = \lim_{X \rightarrow \infty} \Phi_X(A)$ in $\mathcal{H}$ -norm, with the limit existing by Proposition 6.

4 Orthogonal-scattering and intensity

Proposition 8 (Orthogonal scattering). *For every pair of Borel sets $A, B \subset \mathbb{R}$,*

$$\langle \Phi(A), \Phi(B) \rangle_{\mathcal{H}} = \nu(A \cap B), \quad \nu = \delta_{-1} + \delta_{+1}.$$

In particular, if $A \cap B = \emptyset$ then $\langle \Phi(A), \Phi(B) \rangle_{\mathcal{H}} = 0$, so Φ is orthogonally scattered, with intensity measure ν .

Proof. Expand using Definition 7:

$$\langle \Phi(A), \Phi(B) \rangle_{\mathcal{H}} = \mathbf{1}_A(+1)\mathbf{1}_B(+1)\langle \Phi_+, \Phi_+ \rangle + \mathbf{1}_A(-1)\mathbf{1}_B(-1)\langle \Phi_-, \Phi_- \rangle + (\text{cross terms}).$$

Proposition 6 gives $\langle \Phi_{\pm}, \Phi_{\pm} \rangle = 1$ and $\langle \Phi_+, \Phi_- \rangle = 0$, so the expression collapses to $\mathbf{1}_A(+1)\mathbf{1}_B(+1) + \mathbf{1}_A(-1)\mathbf{1}_B(-1) = \nu(A \cap B)$. $\square$

Corollary 9 (σ -additivity and finiteness). *The set function $A \mapsto \Phi(A) \in \mathcal{H}$ is σ -additive in the $\mathcal{H}$ -norm topology and satisfies $\sum_n \|\Phi(A_n)\|_{\mathcal{H}}^2 = \nu(\bigsqcup A_n)$ for every countable disjoint family (A_n) .*

Proof. If $(A_n)_{n \geq 1}$ is a countable disjoint family of Borel sets, only those A_n containing at least one of ± 1 can contribute. Up to relabelling there are at most two such sets, A_{n_-} containing -1 and A_{n_+} containing $+1$, and

$$\Phi\left(\bigsqcup_n A_n\right) = \Phi_- \mathbf{1}_{\{\exists n: -1 \in A_n\}} + \Phi_+ \mathbf{1}_{\{\exists m: +1 \in A_m\}} = \sum_n \Phi(A_n),$$

the series having at most two nonzero terms. The Parseval identity follows from orthonormality of $\{\Phi_-, \Phi_+\}$. $\square$

5 Spectral representation of H : Fubini verification

For $y \in \mathbb{R}$ define the $\mathcal{H}$ -valued integral against a bounded Borel function $f : \mathbb{R} \rightarrow \mathbb{C}$ by

$$\int_{\mathbb{R}} f(\xi) d\Phi(\xi) := f(+1)\Phi_+ + f(-1)\Phi_- \in \mathcal{H}.$$

This is the standard integral against a $\mathcal{H}$ -valued orthogonally-scattered measure supported on the two-point set $\{-1, +1\}$, evaluated as a Riemann sum with exactly two terms.

Theorem 10 (Spectral representation). *For every $y \in \mathbb{R}$,*

$$\widetilde{H}_y = \int_{\mathbb{R}} e^{i\xi y} d\Phi(\xi) = e^{-iy}\Phi_- + e^{+iy}\Phi_+ \quad \text{as an identity in } \mathcal{H}.$$

Proof. We prove the identity by showing that the difference has zero inner product with every generator $\widetilde{H}_s$, $s \in \mathbb{R}$, and then with itself, so that its $\mathcal{H}$ -norm is zero.

Step 1: Explicit computation of $\langle e^{-iy}\Phi_- + e^{+iy}\Phi_+, \widetilde{H}_s \rangle_{\mathcal{H}}$ via Fubini.

Using the Bochner integrals defining $\Phi_{\pm, X}$,

$$\begin{aligned} \langle \Phi_{+, X}, \widetilde{H}_s \rangle_{\mathcal{H}} &= \left\langle \frac{1}{X - X_0} \int_{X_0}^X e^{-ix} \widetilde{H}_x dx, \widetilde{H}_s \right\rangle_{\mathcal{H}} \\ &= \frac{1}{X - X_0} \int_{X_0}^X e^{-ix} \langle \widetilde{H}_x, \widetilde{H}_s \rangle_{\mathcal{H}} dx \quad (\text{Fubini / Bochner linearity}) \\ &= \frac{1}{X - X_0} \int_{X_0}^X e^{-ix} [e^{-i(x-s)} + e^{+i(x-s)}] dx \\ &= \frac{e^{+is}}{X - X_0} \int_{X_0}^X e^{-2ix} dx + \frac{e^{-is}}{X - X_0} \int_{X_0}^X 1 dx \\ &= \frac{e^{+is}}{-2i(X - X_0)} (e^{-2iX} - e^{-2iX_0}) + e^{-is}. \end{aligned}$$

Passing to the limit $X \rightarrow \infty$ (the first term vanishes as $1/(X - X_0)$, and Proposition 6 transports $\Phi_{+,X} \rightarrow \Phi_+$ under the continuous linear functional $\langle \cdot, \widetilde{H_s} \rangle_{\mathcal{H}}$) gives

$$\langle \Phi_+, \widetilde{H_s} \rangle_{\mathcal{H}} = e^{-is}.$$

The analogous computation for $\Phi_{-,X}$ (swap e^{-ix} for e^{+ix} in the kernel; the non-vanishing term is the one with matching phase) yields $\langle \Phi_-, \widetilde{H_s} \rangle_{\mathcal{H}} = e^{+is}$.

Therefore

$$\langle e^{-iy}\Phi_- + e^{+iy}\Phi_+, \widetilde{H_s} \rangle_{\mathcal{H}} = e^{-iy} \cdot e^{+is} + e^{+iy} \cdot e^{-is} = e^{-i(y-s)} + e^{+i(y-s)} = C(y-s).$$

Step 2: Matching inner products. By the definition of the time-average inner product,

$$\langle \widetilde{H_y}, \widetilde{H_s} \rangle_{\mathcal{H}} = C(y-s).$$

So for every $s \in \mathbb{R}$,

$$\langle \widetilde{H_y} - e^{-iy}\Phi_- - e^{+iy}\Phi_+, \widetilde{H_s} \rangle_{\mathcal{H}} = 0.$$

Step 3: Self-inner-product computation. By orthonormality of $\{\Phi_-, \Phi_+\}$ in $\mathcal{H}$,

$$\|e^{-iy}\Phi_- + e^{+iy}\Phi_+\|_{\mathcal{H}}^2 = |e^{-iy}|^2 \|\Phi_-\|^2 + |e^{+iy}|^2 \|\Phi_+\|^2 = 2 = C(0) = \|\widetilde{H_y}\|_{\mathcal{H}}^2.$$

Step 1 applied to $s = y$ gives

$$\langle e^{-iy}\Phi_- + e^{+iy}\Phi_+, \widetilde{H_y} \rangle_{\mathcal{H}} = e^{-iy} \cdot e^{+iy} + e^{+iy} \cdot e^{-iy} = 2,$$

and the conjugate-symmetric inner product satisfies $\langle \widetilde{H_y}, e^{-iy}\Phi_- + e^{+iy}\Phi_+ \rangle_{\mathcal{H}} = \bar{2} = 2$. Therefore

$$\|\widetilde{H_y} - e^{-iy}\Phi_- - e^{+iy}\Phi_+\|_{\mathcal{H}}^2 = \|\widetilde{H_y}\|^2 - 2\operatorname{Re}(2) + \|e^{-iy}\Phi_- + e^{+iy}\Phi_+\|^2 = 2 - 4 + 2 = 0.$$

Hence $\widetilde{H_y} = e^{-iy}\Phi_- + e^{+iy}\Phi_+$ in $\mathcal{H}$, which is exactly $\int_{\mathbb{R}} e^{i\xi y} d\Phi(\xi)$ by the definition of the integral against Φ . $\square$

Corollary 11 (Spectral representation of the Hardy Z -function). *For every $t \geq T_0$,*

$$\widetilde{Z_t} := \sqrt{\theta'(t)} \widetilde{H_{\theta(t)}} = \sqrt{\theta'(t)} \int_{\mathbb{R}} e^{i\xi\theta(t)} d\Phi(\xi) = \sqrt{\theta'(t)} (e^{-i\theta(t)}\Phi_- + e^{+i\theta(t)}\Phi_+)$$

as an identity in $\mathcal{H}$.

Proof. $H(x) = Z(\theta^{-1}(x))/\sqrt{\theta'(\theta^{-1}(x))}$, so $Z(t) = \sqrt{\theta'(t)}H(\theta(t))$, and Theorem 10 at $y = \theta(t)$ gives the stated identity on scaling by $\sqrt{\theta'(t)}$. $\square$

6 Isometry and reality

Proposition 12 (Isometry onto $L^2(\nu)$). *The map*

$$V : L^2(\mathbb{R}, \nu) \longrightarrow \mathcal{H}, \quad V(f) := \int_{\mathbb{R}} f(\xi) d\Phi(\xi) = f(+1)\Phi_+ + f(-1)\Phi_-,$$

is a surjective linear isometry onto $\overline{\operatorname{span}}\{\Phi_+, \Phi_-\} \subseteq \mathcal{H}$. In particular $\dim \mathcal{H} = 2$.

Proof. $L^2(\mathbb{R}, \nu)$ is 2-dimensional with orthonormal basis $\{\mathbf{1}_{\{+1\}}, \mathbf{1}_{\{-1\}}\}$. V sends this basis to $\{\Phi_+, \Phi_-\}$, which is orthonormal in $\mathcal{H}$ by Proposition 6. So V is an isometry onto its image. The image contains every $\widetilde{H}_s = e^{-is}\Phi_- + e^{+is}\Phi_+ = V(e^{-is}\mathbf{1}_{\{-1\}} + e^{+is}\mathbf{1}_{\{+1\}})$ by Theorem 10, so it contains the dense subspace $\mathcal{E}/\mathcal{N}$, hence equals $\mathcal{H}$. $\square$

Corollary 13 (Reality factorization). *Under the isometry V of Proposition 12, the generator $\widetilde{H}_y$ corresponds to the $L^2(\nu)$ -function $\xi \mapsto e^{i\xi y}$, i.e. $V^{-1}(\widetilde{H}_y)(\xi) = e^{i\xi y}$ for $\xi \in \{-1, +1\}$. The two values of this function on the support of ν are e^{+iy} and e^{-iy} , which are complex conjugates of one another. This conjugate symmetry on the ν -support is the precise expression in the spectral representation of the fact that H is real-valued.*

Proof. Immediate from Theorem 10: $V(\xi \mapsto e^{i\xi y}) = e^{-iy}\Phi_- + e^{+iy}\Phi_+ = \widetilde{H}_y$, and $\overline{e^{+iy}} = e^{-iy}$. $\square$

7 Summary

The complex orthogonal measure Φ of the warped Hardy signal H is constructed explicitly by the Cramér-type pre-limit formula of Definition 1:

$$\Phi_{\pm, X} = \frac{1}{X - X_0} \int_{X_0}^X e^{\mp ix} \widetilde{H}_x dx.$$

These are Cauchy in the time-average Hilbert space $\mathcal{H}$ by the direct inner-product computation of Lemma 5 (which reduces to evaluating two one-variable oscillatory integrals, each bounded by 1), and their limits $\Phi_{\pm}$ are orthonormal in $\mathcal{H}$. The limit measure $\Phi(A) = \mathbf{1}_A(+1)\Phi_+ + \mathbf{1}_A(-1)\Phi_-$ is σ -additive, orthogonally scattered, and has intensity $\nu = \delta_{-1} + \delta_{+1}$. Substituting $\Phi_{\pm, X}$ into $\langle \cdot, \widetilde{H}_s \rangle_{\mathcal{H}}$ and applying Fubini/Bochner linearity to interchange the inner product with the Bochner integral gives, after evaluating the two resulting one-variable oscillatory integrals and passing to the limit,

$$\langle \Phi_{\pm}, \widetilde{H}_s \rangle_{\mathcal{H}} = e^{\mp is},$$

and matching inner products against the dense family $\{\widetilde{H}_s\}$ (together with the matching self-norm) closes the spectral representation

$$\widetilde{H}_y = \int_{\mathbb{R}} e^{i\xi y} d\Phi(\xi) = e^{-iy}\Phi_- + e^{+iy}\Phi_+.$$

The Hardy Z -function inherits the representation via the scalar lift $Z(t) = \sqrt{\theta'(t)}H(\theta(t))$.